Aly Maklad

AI Engineer

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SUMMARY

AI Engineer with hands-on experience building agentic LLM applications, RAG pipelines, and multimodal AI systems. Experienced with LangChain, LangGraph, FastAPI, and LLM fine-tuning (QLoRA), with research experience in vision-language modeling and medical imaging. Strong foundation in Python, SQL, and machine learning, with a focus on production-oriented AI systems.

EXPERIENCE

Odoo Backend Developer (Freelance) | BlackBelts ***Aug/2026 – Present | Cairo, Egypt***

- Developed custom **Odoo modules** and **CRM** backend **APIs**, integrating business logic and data workflows to automate operational processes and extend **ERP** capabilities.
- Implemented modular backend functionality and API integrations using **Python**, **Odoo**, and **PostgreSQL** to support scalable business applications.

AI Intern | Digital Hub (D-Hub) & Orange Digital Center ***Jun/2026 – Jul/2026 | Cairo, Egypt***

- Built tool-calling agents (web, math, API) with memory and **RAG** over PDFs and Google Drive using **LangChain** and **Flowise**.
- Designed a **multi-agent system** routing multi-step tasks across retrieval, summarization, and email/report generation.

AI Research Intern | Manipal Institute of Technology ***Jul/2025 – May/2026 | Manipal, India***

- Developed a multimodal pipeline generating radiology reports from 3D chest CT scans using a **vision-language model**.
- Trained a 3D ResNet-50 encoder with MoCo self-supervision, aligning CT features to text via CLIP contrastive learning.
- Built an image-to-text projection module injecting visual tokens into Phi-3, fine-tuned with **4-bit QLoRA** on an RTX 4070.
- Defined evaluation criteria upfront, achieving **0.568 BERTScore-F1** and **0.146 ROUGE-L** on the CT-RATE dataset.

PROJECTS

PropFlow AI | Autonomous Real Estate Lead Management & Sales Automation Platform ***Sep/2026 – Oct/2026***

- Automated end-to-end real estate lead processing using **LangGraph** and **n8n**, transforming incoming inquiries into qualified, scored, and **CRM**-registered leads with automated property matching and personalized follow-ups.
- Streamlined sales operations through **Odoo CRM integration**, implementing round-robin lead assignment, **human-in-the-loop escalation**, and reliable workflow execution using signed webhooks, idempotency, and safe retries.
- Achieved **98–99%** extraction accuracy on a labelled test set and implemented 500+ automated tests, with **CI/CD** deployment through **GitHub Actions** to ensure reliable and maintainable automation.

Khotwa | AI-Powered Adaptive Habit & Productivity System ***Aug/2026 – Sep/2026***

- Built an adaptive productivity system with deterministic scoring, XP, streaks, achievements, and performance analytics, integrated with Google Tasks synchronization, calendar-based reminders, SQLite persistence, and background sync.
- Developed an agentic AI goal planner using **tool calling, Actor–Intervenor workflows, and Reflexion**, combining web research, structured planning, schedule-conflict detection, resource verification, LLM-based critique, and iterative plan refinement.

Multimodal AI Knowledge & Agentic Processing System ***Aug/2026 – Sep/2026***

- Engineered a **multimodal AI system** using **VLMs, ASR, OCR, and LLMs** to reason over video and document content.
- Built a **RAG** pipeline using **BGE-M3 embeddings** and **Qdrant vector search** to retrieve semantically relevant information from processed content, supporting metadata filtering by topics, creators, and other attributes.
- Developed a **conversational AI interface** that enables users to query their personal knowledge base through natural-language prompts, generating context-aware, grounded responses from retrieved evidence.

Sa'ei | AI-Powered Autonomous Job Search & Application System ***Jun/2026 – Jul/2026***

- Built an AI-powered job search platform that discovers and ranks job opportunities, evaluates CV–job compatibility, identifies skill gaps, tailors CVs, and manages application workflows.
- Developed a fully **autonomous multi-agent** pipeline with **LangGraph**, **LLMs**, **tool calling, semantic retrieval, and agent orchestration**, supported by Python, FastAPI, SQLAlchemy/SQLite, and APScheduler for automated execution and persistence.

EDUCATION

Dual Bachelor Degree in Computer Science | Ain Shams University & University of East London ***Sep/2022 – Jul/2026 | Cairo, Egypt***

TECHNICAL SKILLS

AI / GenAI: LangChain, LangGraph, RAG, LLMs, VLMs, Multimodal AI, Fine-Tuning, QLoRA, Prompt Engineering, AI Agents, MCP
ML / DL / CV: PyTorch, TensorFlow, Transformers, Computer Vision, OCR, ASR, NLP, Deep Learning
Programming & APIs: Python, SQL, C++, FastAPI, REST APIs, Webhooks
Databases & Backend: PostgreSQL, SQLite, Redis, Celery, Odoo
DevOps & Tools: Docker, GitHub Actions, CI/CD, Git, GitHub